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1- RUNNING HIGH ORDER SHIM IN CLINICAL APPLICATION

The following section describes the basic routine for in-vivo high order shimming with a patient, and additional parts with a brief explanation of the options available in each screen. Currently there is a specific order of steps that need to be followed to use the shim process in regards to the shim series capturing any S/I FOV offset that may occur between the localizer series and subsequent clinical series.

1-1 Process

The only recommended process for using this feature is as follows:

1. Prescribe and run a localizer series as is normally performed.

2. Prescribe and run any other series that you wish to run prior to specific localized shimming or just go right to the shim series.

3. Prescribe and save the clinical series you wish to run following the shim series before pulling in the shim series. Remember to turn off the “Autoshim” option in this series since the High Order Shim will take precedence. Make sure there is a series description for each series used. The High Order Shim series will use the series description as shown in step 6. (Further notes regarding the protocol Autoshim button are found below in the section “Autoshim Button”).

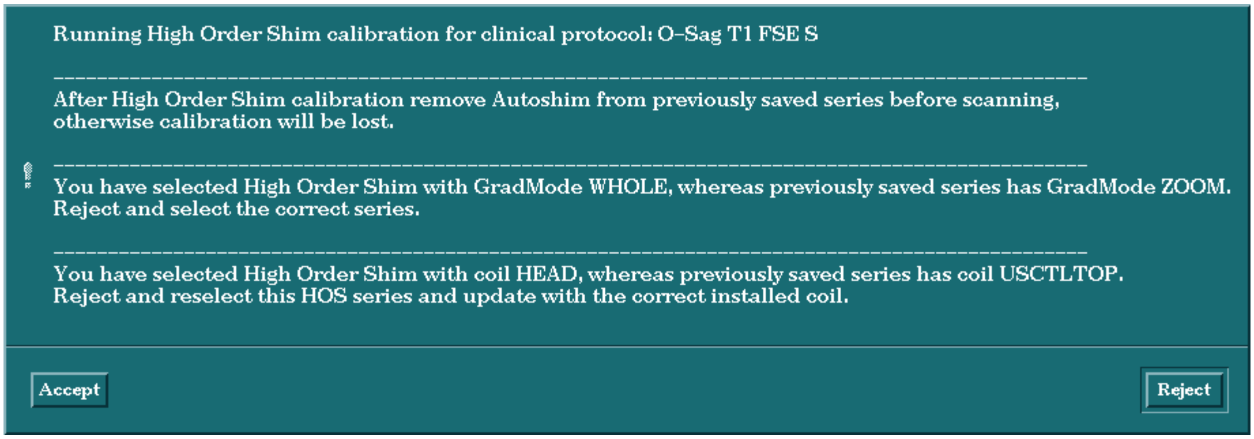
4. Choose one of the series from the **GE/other/HighOrderShim** protocol. The choice depends on the expected maximum FOV (some possible FOV’s depending on system type are 24, 28 and 48) and the Gradient coil, if TRM, (**WB** for WHOLE-BODY or **ZM** for ZOOM) to be used during clinical scanning. (The shim series are further described below in the section “Available Shim Series”). Change the patient position parameters and RF coil to match those of the clinical series to be run next. This HOS scan should be run just prior to the actual clinical series, but after having used “Saved Series” for the clinical series. A change in the Gradient coil used and/or cradle location for other series with the same patient will require a rerun of the HOS scan.

5. Save the series without changing any of the scan parameters other than those mentioned in the previous step.

6. At this point the shim series should give you the following message:
   “Running High Order Shim calibration for clinical protocol: Saved *series description here*”
   Other messages will also appear in the same message pop up window if there is any conflict identified between the clinical series and the shim series such as the GradMode (for TRMs) and RF coil used do not match the subsequent series to be scanned. See Illustration 1 for messages that may appear.
   If this information shows a problem then pull in a new shim series and make the appropriate modifications. The system will now ask you to press advance to scan so it can move the table to the new position.

7. Run the shim series, and complete the HOS calibration as described in section 1-4 on “the Shimming process.”

8. Prep and run the previously saved clinical series. No cradle movement should occur at this time. If you did not deselect Autoshim previously, use view/edit to deselect autoshim, Save Series and proceed with the scan.



Potential Messages seen from a saved shim series
Illustration 1

1-2 Available shim series

There may be up to four shim series available defined by FOV and GradMode (for TRM's). The larger FOV series is primarily used in the body coil but can also be used for receive only surface coils and for small volume T/R coils that are off-center such as the wrist or knee coils. For use with the volume T/R coils, you may want to update the coil used after selecting the head shim series, to benefit from the increased SNR.

The smaller FOV series is used primarily for the head coil and can also be used for the knee T/R volume coil if the knee is positioned at isocenter. Anything outside the smaller FOV region at isocenter must be shimmed with the larger FOV series.

To use the smaller FOV shim series for a coil other than the head coil, select the series matching the GradMode you will use in the next clinical series and then change the patient orientation and coil name to match.

Choose the GradMode to match with that defined in the clinical series just saved. In the event of a mis-match, a message to this effect will be given upon Save Series of the HOS series.

1-3 Autoshim Button

Anytime you enable the "Autoshim" button after running a High Order Shim series, you will see the following message:

"You have set ON the autoshim button after having run the high order shim calibration.
If this remains ON, scanning with this series will reset ALL the shim coils."

This is telling you that all the shims channels will be reset to zero as if you had never run the shim series. Reselecting Autoshim should be done anytime you change locations in the anatomy that are outside the previously shimmed volume or that will cause the table to move due to a new S/I offset. A new shim series should also be performed for any new slices prescribed that will fall outside the previously selected shim ROI even if an S/I table movement is not required. This is due to the fact that ONLY the region inside the ROI used in the last shim series will be shimmed as noted in the section "The Shimming process".

1-4 Shimming process

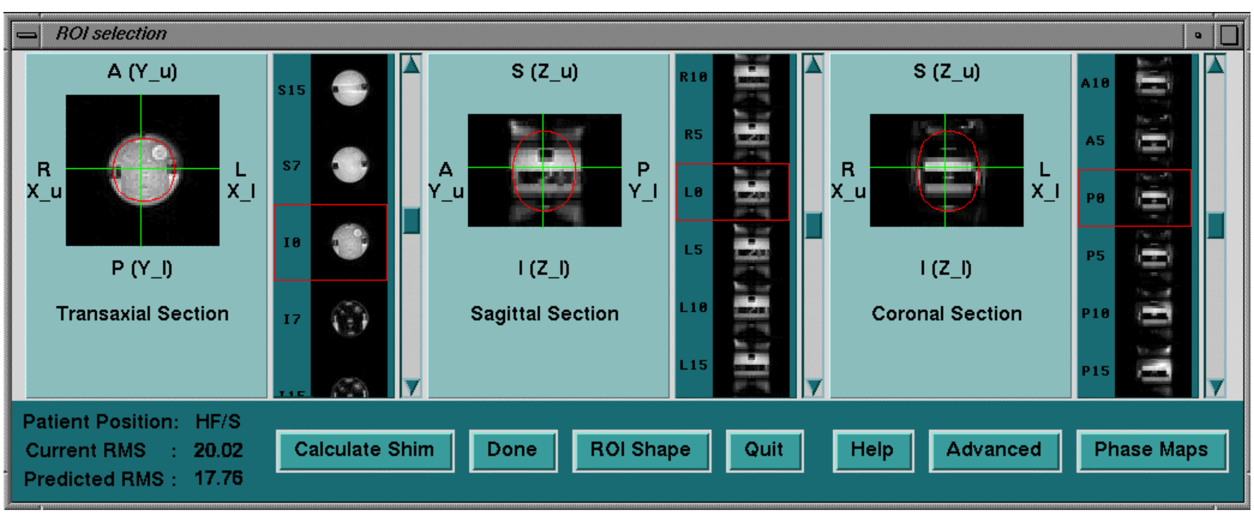
After the shim scan and reconstruction finishes, the Shim User Interface will appear as shown in Illustration 2.

Note

The processing and displaying times of this interface are a function of the HOS series used, and may take many seconds to appear. The smaller FOV series processing will only take 5-10 seconds whereas the larger FOV series will take 30-45 seconds before the shim interface will appear.

The Shim User Interface is where the action of defining a specific region to shim takes place. Only the region inside the ROI used in the shim series will be shimmed, given that the patient has not been moved. Anatomy outside the region selected for shimming may be adversely affected by the shim process and may cause image distortions and artifacts. The High Order Shim process is for improving localized areas of anatomy to correct for affects of the patient in the magnetic field. This process improves further upon the capabilities of the current Autoshim process but only in the area selected by the ROI in the shim user interface. The shim of the regions surrounding the ROI is sacrificed in order to allow the area inside the ROI to be further improved. It is important to insure that the ROI selected will cover the slices desired in the following series.

Selecting a region of interest (ROI)



Magnitude Images in the basic screen
Illustration 2

For selecting the ROI, the operator can select any slice from the corresponding scrollbar in each direction. The slice may also be selected by using the two green intersection lines on each of the three images (Axial, Sagittal & Coronal). The selected ROI may be either rectangular or elliptical.

The operator may drag and resize the ROI on any image (details in Help). The ROI displayed on the other two images will be updated accordingly.

In the basic screen, the user may select any of the following features, by clicking on the appropriate button:

- **ROI Shape** - to select between rectangular and oval shape.
- **Green lines** - to off-center the intersection between all three displayed images.
- **Calculate Shim** - to run the shim values calculation with the current ROI selection.
- **Done** - to download the calculated values to the Shim supply and exit.
- **Quit** - to exit without download.
- **Help** - has instructions for manipulating the ROI and the green lines, as follows:

*** Change ROI shape by clicking on 'ROI Shape' button , select either "rectangle" or "oval"**
*** To move the ROI, place cursor inside the ROI and drag**
*** To resize the ROI, place cursor outside the ROI and drag**
*** To move the intersecting planes one by one, place cursor on one of the green lines outside the ROI and drag**

- **Phase / Magnitude Maps** - to toggle between displaying phase or magnitude images.The Phase Maps can help identify the location of off-center anatomy such as knees, wrists or shoulders. Small regions will show up as small objects and hard to see in the large FOV shim series. The phase maps can help in locating a suitable ROI around the anatomy.
- **Advanced** - to open the ADVANCED window for additional choices, see below for details.

a. Manipulate the ROI on the images so that it occupies the desired region.
b. Click on **Calculate Shim**. Now the optimum values to improve the magnet homogeneity within the specified ROI volume are calculated. The results are shown as the current and predicted RMS as shown in Illustration 2. See “Shim Improvement identification” for more details including information regarding Section 3, High Order Shim Failures.
c. When the calculation is complete, the Done button will highlight. Any ROI size/position changes and subsequent “Calculate Shim” requests must be done prior to selecting the Done button.
d. Click on **Done**. At this point, the new values for the central frequency and linear shims are transferred to the system, and the five high order shim currents are downloaded to the shim supply channel by channel.
e. Once the download is complete, the High Order Shim interface will close.
f. If required, run the shim scan again by selecting “Scan” to check the shim corrections. Ensure prior to any scanning that the HOS screens have been closed (the HOS screens remain open after download when this action is run from the Advanced screen).
g. When the User Interface comes up again, just select “Calculate Shim” and verify that the current RMS value has improved. See the section “Shim improvement identification” for details.

1-5 Shim improvement identification

To identify the improvement in the shim of the ROI selected look at the RMS deviation current and predicted fields in the Shim UIF after positioning the ROI and selecting “Calculate Shim”.

The current field tells you what the shim is now and the predicted will give you a sense of how much it might improve after Done/Download is run. Any shifting by the patient will make the results shift. Generally if the predicted value is close to the current value(within 1-2Hz) then there will be very little benefit in running another iteration. Generally only one or two iterations should be necessary.

If there seems to be little to no improvement for the very first pass, select the Advanced button to open the Advanced screen and make sure that all shim coils are enabled for use on the right hand side. This will be evident by see that the buttons next to each coil is indented (selected), if not, then select them all and calculate shim again. They will stay enabled until someone manually disables them. See Illustration 3 for the Advanced screen and coil buttons.

2- HIGH ORDER SHIM SOFTWARE

There is no Window/Level capability in the shim UIF. If the images come out very dark the shim calculation will fail since it thresholds the data and will not use anything under a certain pixel intensity. This is not configurable and there is nothing you can currently do about it.

Basic use of shim interface is to run the shim scan, position and size the ROI on the image, select "Calculate Shim", look at the current and predicted shim, select "Done" to download and exit and rescan to check shim improvement or just go to next clinical series.

The High Order Shim software is contained mainly in two directories. All of the scripts and executables are contained in the /usr/g/bin directory. The calibration files and other data files are in /w/config/ho_shim.

2-1 Commands

The commands and user interface software are all in the /usr/g/bin directory so commands such as setcurrent, readcurrent and setzero can be run from anywhere.
The full list of commands is as follows:

- **setzero** – download to all high order shim (HOS) channels in the supply, the value zero.
- **readallcurrent** – display the current value of the linear shims being used (no reference made to the Gradient coil currently active) and the current value of the high order shim channels read directly from the Shim supply. The returned value may be in Amps or Dac units depending on the system you are on. If the value is returned in Dac Units, the conversion is 8192 dacs = 1 Amp.
- **setCurrentLinearShim aa bb cc** – update the linear shim value for X=**aa**, Y=**bb** and Z=**cc**. The Gradient coil to which these shim current values are appropriate corresponds to the previous coil used in the HOS Autoshim scan.
- **readCurrentLinearShim –** display the current values being used for X, Y and Z.
- **setcurrent** – used to set a current in the shim supply. All software releases for all platforms use Amps as the value parameter for the setcurrent command such as: setcurrent xy 1 (sets 1 amp in the xy channel)
- **readcurrent** – used to read a specified shim power supply channel. I.e. readcurrent z2

2-2 Calibration files

The files generated during the HOS Field Map Creation procedure tell the High Order shim software what corrections to make for what field disturbance seem in a manner similar to LVShim. The reference maps are the cal files that are generated by the "makereference" script and are kept in /w/config/ho_shim/reference.
They will all have names such as:
HO_MAP_GRD_W.FOV_24_24_18.RES_64_64_32
The "W" is for Whole GradMode, FOV is 24cm, with a 64 squared resolution. There will be another file that has a "Z" instead of the "W" which is for GradMode of Zoom. If you are on a 3T94 with CRM gradient, then you will see a "C" for CRM. You may also see an FOV of 28 instead of 24 such as: FOV_28_28_21
There will also be two files for a 48cm FOV with a resolution of 128 squared.

Along with each of these files is a file with the same base name but has ".params" appended. This is a file that defines what current offsets and timing parameters were used for each reference map. Below is a sample params file:

Sample params file

```
# size of reference maps
64 64 32

# phase delay
2

# number of shims
8

# shim characteristics:
# name delta units active
x       10      gradunits   1
y       10      gradunits   1
z       10      gradunits   1
zy      0.5     amps        1
x2y2    0.5     amps        1
zx      0.5     amps        1
xy      0.5     amps        1
z2      0.25    amps        1
```

The last section of the file shown above shows the delta current offsets used for each channel during the reference map generation. In this case 10 gradunits offset was used and 0.5 amps were used for most channels except Z2. There is also a column shown called “active”. This column will show a “1” for all channels that are enabled via the shim user interface advanced screen. It will show a “0” for any channel disabled during the last shim run. **Refer to the Advanced screen display in section 4-1 Illustration 3 to see the enable/disable buttons.**

Another file seen for some systems (eventually all systems) is a mask file that has the same base name as stated above but ends in “.mask”. This file is a mask applied to the reference map data to “mask out” the background noise and any unusable data such that only valid data is used for shim calculations.

Also seen in /w/config/ho_shim/reference are two files that are links:
reference.field and reference.params
The system will change these links to point to the specific HO… reference map file being used for the current shim iteration. The scan software changes this link depending on Gradmode and FOV used in the shim series.

2-3 Library files

The files saved when using the Library screen from the shim user interface are kept in:
/w/config/ho_shim/mr_hshim_config.WHOLE (for TRM gradient systems)
/w/config/ho_shim/mr_hshim_config.ZOOM (for TRM gradient systems)
/w/config/ho_shim/mr_hshim_config (for 3T94 CRM systems)

There is also a link for TRM systems named: mr_hshim_config that is used to point to which library directory is currently in use per the GradMode selected for TRM gradient systems.
The files are named “hoshim01” through “hoshim10” which are pointers to the files also kept in these directories that have the saved shim information. The files with the shim information use the naming convention: hoshim01.howgood
These are text files that can be looked at via “**more hoshim01.howgood**” and you will see

Sample howgood file:

```
Date: Wed Nov 14 06:59:54 CST 2001      Run Number:  11776
==================================================================

======================================
Force Patient position Head First/ Supine
... forced mapentry is 1
... forced maprot is 0
W:      106.67     2.9822     18.609     16.614     17.786     12.648     11.133     10.175     10.56

condition number of matrix  =   35.76746

condition threshold         =   50.00000

      Shim
============================

Field Map Information
---------------------
File ......... data/P99999.7
Size ......... 64 64 32
Phasedelay ........ 4.48

Reference File Information
--------------------------
File ..............
/w/config/ho_shim/reference/HO_MAP_GRD_W.FOV_32_32_24.RES_64_64_32.params
Size .............. 64 64 32
Phasedelay ........ 2.00
Number of shims ... 8

Shim Volume Information
-----------------------
Shim region: elliptical
Shim performed over x=13,-15, y=13,-15, z=10,-9
Oblique angle shimmed over:  0.00
Number of points in VOI .............. 7666
Number of shims used for solution .... 8

Current Field Map Statistics
```

```
----------------------------
RMS deviation ....   3.46 Hz
Peak deviation ...  23.00 Hz

Predicted Field Map Statistics
------------------------------
RMS deviation ....   1.04 Hz
Peak deviation ...  10.55 Hz

Delta currents required
-----------------------
z0                        ... 2.794 Hertz
x                         ... 0.240 gradunits
y                         ... 1.819 gradunits
z                         ... -0.127 gradunits
zy                        ... 0.086 amps
x2y2                      ... -0.015 amps
zx                        ... 0.033 amps
xy                        ... 0.036 amps
z2                        ... 0.004 amps
Coils absolute values
=====================

R_Z0 63859957
R_X   1
R_Y   4
R_Z   -1
R_ZY 0
R_X2Y2 0
R_ZX 0
R_XY 0
R_Z2 0
```

This file shows all the information specific to the saved shim values. The particular reference map and ROI placement information is stored along with all the other shim information shown at the time of the shim calculation. This is the same information displayed in the Advanced screen output window with the exception of the very last section for Coils absolute values.

2-4 Restore Defaults

The Restore Defaults button in the Advanced Display will take the “Coils Absolute values” listed in the file “hoshim01.howgood” and compare them to what the values currently in the shim channels. The software will then insert into the Delta current fields (on the Advanced screen) the changes necessary to get from the present currents to the default currents. Since the default currents in the howshim01.howgood files should be zero, the delta currents should just be opposite polarity of the present shim coil settings.

2-5 HOS Field Map Creation:

There are two main scripts that are run during initial field map generation.

The first is the “create_howgood_lib” that generates the files in the Library directory spoken of in the section “Library files”. This is only run once per software load. If you run it when the library directories already exist, the script will tell you they already exist.

The makereference script is the script that calls all other scripts necessary to generate the reference maps. This script is the one that asks the questions for which FOV and gradient mode you wish to generate a reference map for. It then calls the build_ref_acq script that goes on to scan a baseline and then run all the scans with one shim coil offset per scan. From this data, the special reconstruction software generates the difference maps for each of the shim channels and concatenates all the information into one reference map file for later use. The shim scan used is the same spiral PSD with the same parameters as used during clinical shim series.

During the makereference execution, you can see the output from the scripts for each scan run. The sequence is as follows:
Scan 1: Baseline using Autoshim to define the starting gradshims.
Scan 2: X
Scan 3: Y
Scan 4: Z
Scan 5 : ZY
Scan 6 : X2-Y2
Scan 7 : ZX
Scan 8 : XY
Scan 9: Z3 (only exists for 3T94 magnets)
Scan 10 : Z2
During the scans for the gradshims, you can see in the script output the values for the baseline gradshims and then for each of the 3 gradshims, the offset being applied. For instance, if the baseline gradshims were –2, 2, 1 (X,Y,Z) then the X scan will show the gradshims set to: 8,2,1 (assuming a 10 offset for 24cm fov).

It is a good idea to pay attention to the screen as it runs the scan to make sure you don’t see any MXA communication time out errors. If the communications to the shim supply fail at any point during the reference map generation, the resultant reference map will be corrupted. There currently are no checks for communication failures during this process.

The delta currents used for generation of the reference maps are different for each FOV since the sensitivity by FOV and phantom used is different. They are optimized for best response without causing phase wraps in the desired FOV.

2-6 Log File

/usr/g/service/log/ho.log – to view the results of all the calculated shims iterations performed on the same patient. The latest one run is at the **top** of the file.

2-7 Configuration File

/w/config/prescan_config.cfg – This config file is the only system configuration file that contains a High Order Shim entry. At the very bottom of this file is a line: hos_exists = 1 (or high_order_shim_coils_exist for 3T94)(1 = shim option installed, 0 = no shim option). The shim software triggers off this entry to enable or disable code specific to the shim option. Shim currents **will not** be auto reset between exams or when autoshim is selected after hoshim series if the prescan config flag above is set to zero.

2-8 Calling up the Shim interface

The Shim user interface can be called up again after a shim iteration and prior to the next series by opening a C-Shell and typing "**recon99**". This is the script that is called by scan after the shim series is scanned. This can be seen within the shim series by looking at the CV page and seeing that the recon entry is set to 99.
This command can be used if you closed the shim interface accidentally or wish to restart if for some reason.

Note

Running a shim scan back to back without exiting the shim interface will cause multiple interface windows to open. The same is true when using recon99.

2-9 Shim Interface ROI settings

After sizing/positioning the ROI in the shim interface, the "Calculate Shim" button is generally used to find the shim inside the ROI. The information for the location/size of the ROI is listed in the Advanced screen output window as follows:

Shim Volume Information

Shim region: elliptical
Shim performed over x=13,-15, y=13,-15, z=10,-9
Oblique angle shimmed over: 0.00
Number of points in VOI 7666
Number of shims used for solution 8

The line "Shim performed over" shows the coordinates of the ROI. These are also stored in a file: /w/config/ho_shim/tk.argument.64 or .32 depending on the resolution. The .64 file is for the 48cm FOV maps and the .32 is for the 24cm FOV maps.
These are text files that can be viewed to see what was last used as coordinates for the ROI.
There is no easy way to know from these coordinates exactly how to position the ROI to match other than estimating the size in the interface. The image sizing in the interface is as follows:
24cm FOV: X = 64, Y = 64, Z = 32
48cm FOV : X = 128, Y = 128, Z = 64
The window is split in half, so for the 24cm FOV X direction, the size is +32 to –32. From this you can estimate where to place an ROI, but will need to use "Calculate Shim" and look at the Advanced window output screen to see what the size/position of your ROI actually is.
For general use this is not needed, since the very last used ROI size/position is recalled for the next shim iteration or series.

3- HIGH ORDER SHIM FAILURES

If the shim calculation fails you will see an error message pop up stating: “SVD error: not enough points”. This is telling you that the software could not identify enough data points within the ROI that are above a certain pixel intensity threshold to be used in the calculation of the new shim settings.

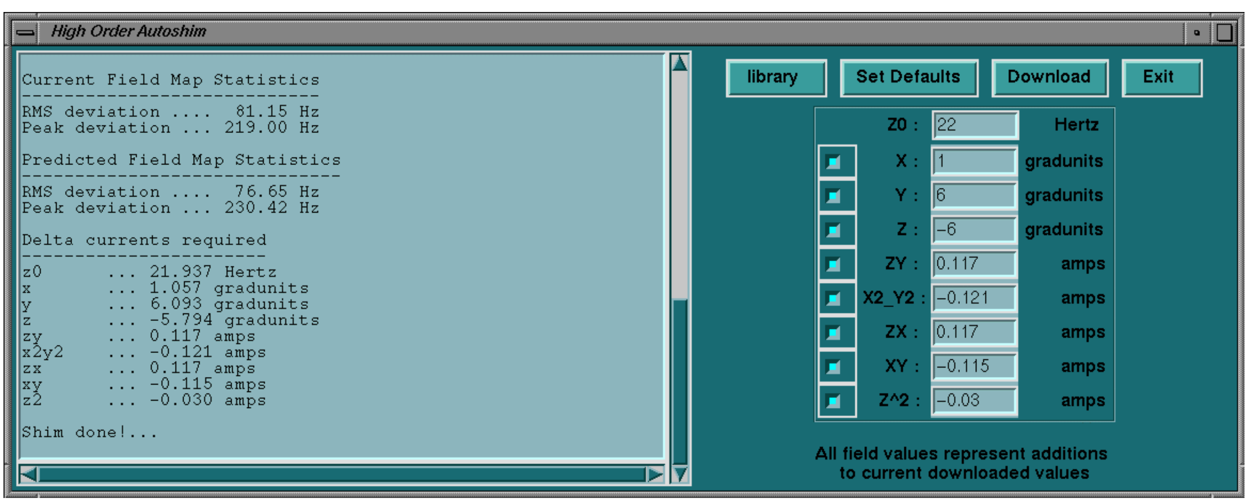
Possible causes:

- **The image is very dark. i.e. not enough signal.** It may be possible to fix this by exiting the shim interface, running autoprescan again, enter manual prescan and increase the R1 receive gain by 1 or more counts while looking at the Scan TR entry point. May also use the R2 gain if R1 is at maximum. If this does not help, then the anatomy currently being imaged will not be able to be shimmed.
- **The ROI contains too much of the background inside the ROI boundaries**. Check the ROI size and position, shifting and repositioning as necessary. Selecting Calculate shim repeatedly during this process will let you know when the problem is corrected sufficiently to provide shim information.
- **The patient is causing a severe shim distortion.** This may occur for many reasons such as imaging near metal implants that cause RF reflection. Selecting “Phase Maps” from the shim interface can help identify this issue. What you will see if there are severe shim distortions are white and black stripes side by side inside the ROI. The shim option is not able to correct for shim distortions that are this severe.

4- ADVANCED SCREENS

4-1 Advanced Screen

1. To review details of the shim calculation results, save a shim-set of data, recall a previously saved set, manually enter values or remove one or more of the shim channels, click on **Advanced** in the basic screen, as seen in Illustration 3.
2. The Advanced screen as seen in Illustration 3 appears.



Advanced screen
Illustration 3

In the Advanced screen, the user may select any of the following features by clicking on the appropriate button:

- **Log window** – to review a log of the shim calculation and download details. An estimate as to the quality of the improvement after the channels are updated can be seen by looking at the change in RMS deviation between the Current and Predicted Map statistics.
- **Library –** to open the LIBRARY window, to either save a shim-set or recall a previously saved one. See below for details.
- **Set defaults –** to select a set of predefined shim values, corresponding to resetting the linear shims to the original system LVShim values, and with high order shim channels set to zero.
- **Download** – to load the current set of shim values to the Shim supply.
- **Exit** – to return to the basic screen window.
- **ON/OFF for each shim [**small button alongside each channel, default is **ON]**– to remove a linear and/or high order shim channel from the current shim set, i.e. not to use it for updating the calculation or downloading to the Shim supply. More than one can be removed for this purpose.
- **Field** – difference value to be added to this channel, entered automatically after use of **Calculate Shim**, or manually using this screen.
- **Field range of value** – in table 3-1. These are suggested ranges for each channel based on expected changes in field homogeneity, and are not necessarily typical values.

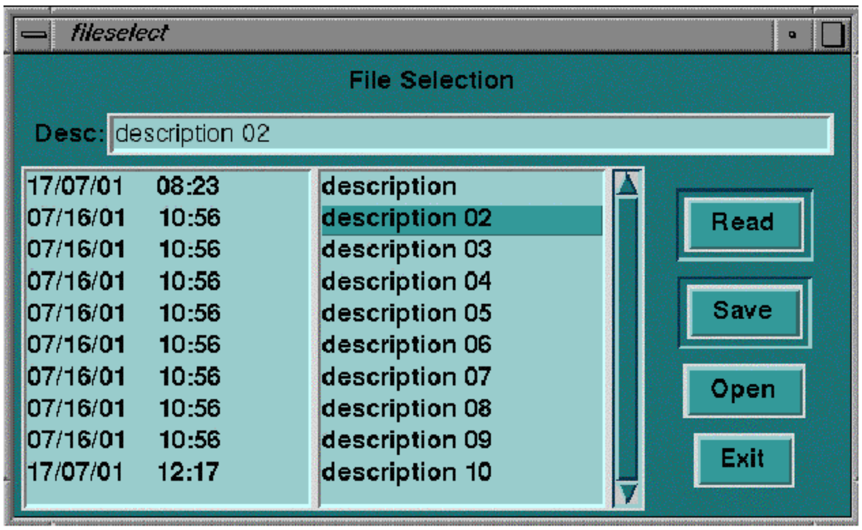
The numbers shown in the advanced screen for each channel are add-on values to whatever is already in that channel. The ranges shown in the following table are guidelines for the largest typical offsets requested for any single iteration. Values less than +/- 0.010A are considered negligible. Significantly higher may indicate a problem. See section 5 for troubleshooting.

Table 3-1
Suggested range of maximum values for each channel

Channel	Units	Negative limit	Positive limit
Zo	frequency in Hertz	-500	+500
X	linear shim in gradunits	-20	+20
Y	linear shim in gradunits	-20	+20
Z	linear shim in gradunits	-20	+20
ZY	**high order shim in A**	- 0.500	+ 0.500
$X^2\_Y^2$	**high order shim in A**	- 0.500	+ 0.500
ZX	**high order shim in A**	- 0.500	+ 0.500
XY	**high order shim in A**	- 0.500	+ 0.500
Z^2	**high order shim in A**	- 0.250	+ 0.250

4-2- Library screen

1. To save a shim-set of data, recall a previously saved set, or manually adjust a previously saved set, click on **Library** in the Advanced screen, as seen in Illustration 3.
2. The Library screen as seen in Illustration 4 appears.



Library screen
Illustration 4

In the Library screen, the user may select any of the following features by clicking on the appropriate button:

- **Desc** – used to give the new file a description, which will appear in the list in place of the current ***description*** line. The first line is reserved for the default linear shim values.
- **Read** - used to read the existing set of shim values from the Shim supply and grad shims, and present the difference between these and the highlighted set in the Library in the Advanced screen.
- **Save** - used to save the currently displayed set of shim values to the selected description file.
- **Open** – used to open an editor with a previously saved set of shim values, which are afterwards read in as the difference between these values and the currently downloaded set.
- **Exit** – used to return to the Advanced screen.

5- HIGH ORDER SHIM TROUBLESHOOTING

The following flowchart is for high order shim troubleshooting:

Problem: Hoshim does not work

Prerequites for Hoshim:
Stable system – SPT passes.
Product spiral images look fine.
System Base shim is in Spec.

Check to see if the hoshim series will download and scan.

Pull up the shim series from GE/other, save and and scan the series. For head coil, use 170mm ball without loader on a pad. For body coil, use 270mm body SPT ball without loader.

Does error log show download failures.

Yes

If the shim series fails to download, the system may not be finding the reference maps. Compare file name in error log to files found in /w/config/ho_shim/reference.

Are the images in the shim GUI distorted or see image anomolies?

Yes

Look for dark bands in the sagital and coronal images in the shim GUI. If seen, then psdRfWait is wrong. Run a product Spiral GRE scan. If this is also distorted then a system problem exists that must be resolved first. Look for corduroy in images and if seen, troubleshoot white pixels.

Does the shim calculation fail?

Yes

Make sure the ROI is centered in the phantom and covers approx. 75% of phantom volume. If it still fails, reference maps may be corrupted. See section 1-4 on troubleshooting reference maps.

Are there communication failures to shim supply in error log?

Yes

See communication troubleshooting section 5-3.

Is the Predicted RMS shim worse than the Current RMS shim?

Yes

Check the Lvshim gradshim to verify base shim of system is in spec. If base shim is ok, then suspect corrupted reference maps.

Does functional test of reference maps from calibration procedure fail?

Yes

Suspect corrupted reference maps. Regenerate reference map that failed and try again.

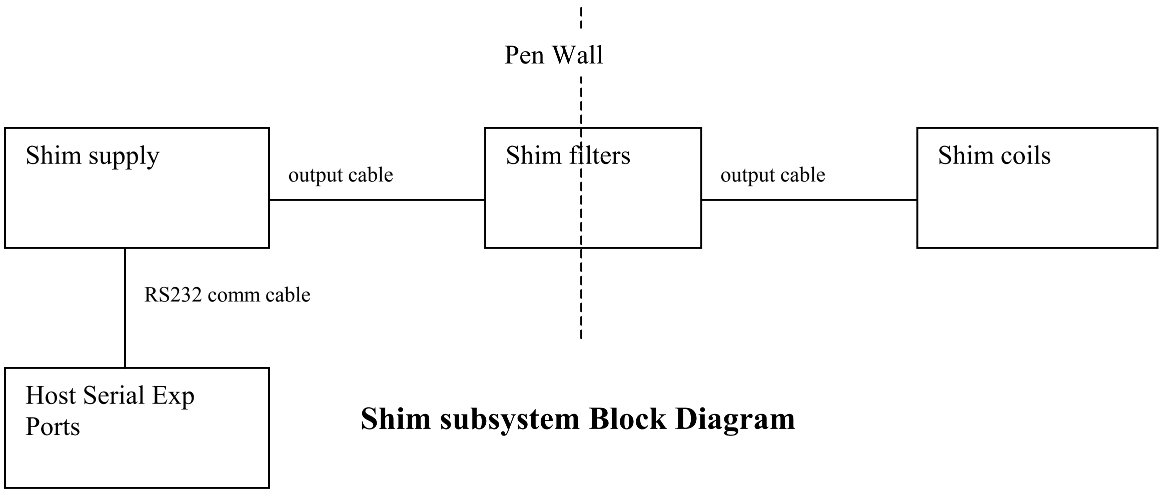
Does one or more shim channel seem to not work from functional check?

Yes

Manually set currents into each shim channel and verify output via front display or multimeter connected to banana jacks on supply. See shim hardware troubleshooting section 5-1.

5-1 Hardware Troubleshooting

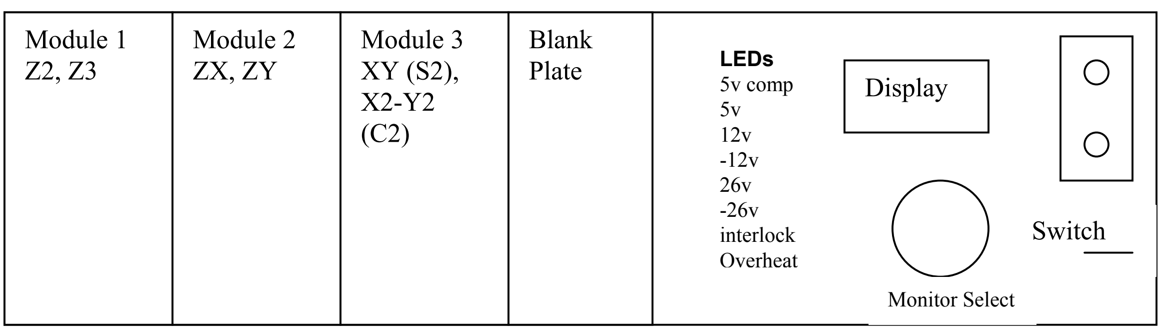
The included block diagram shows the basic hardware components of the high order shim subsystem.



Shim subsystem Block Diagram

5-1-1 Components

MXA-6 Resistive Shim Supply:
Shim supply output modules are all fully interchangeable. There are 3 dual channel modules. Looking at the front you will see the following layout.



The LEDs show the status of the power supplies, communication interlock and overheat fault. All but the overheat LED is green. During power up there is a short delay prior to the interlock LED lighting up. The supply will only communicate with the host after the interlock LED is lit. The display is actually part of a voltmeter unit in the supply that looks at the power supplies and current output as converted to a voltage. There are two banana jack ports on the far right that can be used to check only the current output (as read in mV not amps). The conversion as printed on the front of the supply is 100mV per Amp. The Switch changes the meter display and Monitor Select dial to either power supplies or current outputs (V mode or I mode). The banana jacks only give the current output with the switch in the “I mode”, at which point the dial selector is used to define which shim channel output you are looking at.

For TRM’s, there is no Z3 channel, so expect to see zero on that channel at all times.

All three modules are identical and interchangeable. There are no settings so modules may be swapped by just turning off the supply, swap modules and turn it back on. This is an easy way to identify a bad output module to confirm the problem is internal to the shim supply.

Display on Shim Supply:
The digital display on older units has been known to burn out, to which there is a modification going into all new supplies to try to eliminate this issue. This does not affect the operation of the supply only the ability to look at the power supplies. Can still read current output via banana jacks on the front of the unit using the dial to select the output channel. Keep in mind the conversion of 100mv=1Amp that is printed next to the banana jacks.
This display is part of an internal multimeter connected to the outputs of the shim channels. Expect to see displayed values fluctuating during scans. If the values are not stable when no scanning is in progress, there may be a problem. Use a multimeter connected to the banana jacks to confirm a problem on the shim output vs. display failure.

Shim Output Cables:
The output cable pinout for the MXA-6 6 channel shim supply is as shown below. This is valid for all TRM gradient systems with the high order shim supply. 3T94 Eclipse systems may have an 8 channel shim supply in the near future that will have a modified pinout.

COIL	OUTPUT	RETURN	Nominal Resistance (3T94)	Nominal Resistance (TRM)
Z2	5	6	5.7	3.6
Z3	7	8	5.5	N/A
ZX	17	18	2.9	3.5
ZY	19	20	2.9	3.6
X2-Y2	25	26	3.9	4.3
XY	27	28	3.9	4.3
GROUND		37		

Pen wall filter:
The internal labeling of the pen wall shim filter is shown as supply 1 thru 6 and follows the dial order of channels in that supply 1 is Z2, 2 is Z3, 3 is ZX, etc.
FE's can look for oscillation on a channel via differential connection across the two filters for any one channel. May need to pull back heat shrink covering on terminals.
Each shim channel pair of wires goes through individual pi filters that have capacitors to ground on each end of an inductor. You should not be able to measure a short to ground from the filter center pin to the pen panel. If you can measure a short to ground, then a replacement pen panel filter assembly is required.

5-2 Communication problems

The communication cable from the supply J1 to Serial Exp port 2 on the host is RS232 and should be no more than 50 feet per RS232 standards. The supplied cable is 16m which is 52 feet. If an extension cable is added, communication errors can result depending on the quality of the cabling and the connection. It is believed that the extra impedance in the connectors is the cause of communication errors.

This will show up as “MXA communication faults” that may not recover. They may require a reset of power to the shim supply.

To check communication between the Octane computer and the Shim power supply it is possible to send a current command from the Octane computer to the Shim power supply and then to read the current back into the Octane computer.

- The shim power supply is capable of setting output current to the shim coil from a range of –4 Amp to + 4 Amp.
- There are five shim coils that are defined in the computer as shown in table 5-1.

Table 5-1
Shim coil definition

Shim coil name	ZY	ZX	XY	X^2-Y^2	Z^2
Shim supply name	zy	zx	s2	c2	z2

A typical command to set current and to read current from the shim power supply is shown below:

For all systems:

- Open a C-shell window and type:
- **setcurrent aa bbbb <enter>**

(**aa** defines the Shim coil name of the appropriate shim coil, **bbbb** defines the desired current in Amps)
i.e. setcurrent X2Y2 0.25

For 3T/94:
The value returned is in shim units, 1000 mA = 8192.

- **readcurrent aa<enter>**

i.e readcurrent X2Y2, Response = 2048 shim units.

For Twinspeed:
The value returned is in Amps.

- **readcurrent X2Y2 <enter>**

Response = 0.25 Amps

If the computer prompts back with the appropriate response that means communication with the shim power supply is OK, otherwise check for the correct connection, and or proper operation of the shim power supply.

Attempts to set any channel over 4 amps will cause an “over margin” error to be returned. Currents can be manually set via “setcurrent” and read using “readcurrent”. There is also a “readallcurrent” command to read the gradshims and all 5 second order shim channels.

Check to make sure that the cable from port 2 of the host serial expansion port is connected to com1 on the rear of the shim power supply.
Make sure the Interlock light on the front of the power supply is lit. If not lit, then cycle power and wait for the power up cycle to complete. If it still fails to light then there is a problem in the power supply.

5-3 Troubleshooting a communication problem

A communication problem with the shim supply can be troubleshot using the “cu” command and/or FE laptop.
There are two ports on the rear of the shim supply, a diagnostic port (J2) and serial com port (J1).

The communication setup for using Hyperterminal on the laptop is:
19200 baud, 8 bits, no parity, 1 start/stop bit, no flow control

Connect a serial cable from your laptop serial port to the J1 connector on the rear of the shim supply. Power up the shim supply and you should see a “.” (period) prompt after the shim supply has completed initialization indicated by the “interlock” led lit on the front of the supply. At this point you can use the following shim commands to directly talk to the shim supply:

.sc xy 8192	This is set current for the xy channel to 8192 dac units which is 1 amp.
.gc xy	This is get current for the xy channel for which the response will be 8192 after running the sc command shown.

If these commands function for the shim channels, then the supply communication is good.

Connecting to J2 will show the power up messages and runtime messages after powering on the shim supply.

To test communications from the host, the “cu” command can be used from a Cshell window as follows:
cu shimsupply

This will connect to the shim supply through the serial expansion module giving you the same command functionality as described above for the laptop using the sc and gc commands. Use “~.” (tilde period) to exit this mode. Suggest also using “cu ipg” or “cu tps” to talk to the tps or ipg to insure the serial expansion module is functioning for service communication paths to the tps and ipg. Refer to the “TPS/ISE Serial debug connection” document SC1TSC3.doc found in the LX service methods under System cabinet troubleshooting section. A quick check looking for serial connections to the expansion module ports 6,7,8 will tell you if the cabling is connected for talking to the ipg and TPS.
If you are unable to talk to the tps or ipg then a power reset to the serial exp. Module and full host reboot may be necessary. Troubleshoot the tps/ipg communication via the above mentioned document prior to trying the shim supply again.

5-4 Troubleshooting potential interaction between the shim supply and IQ issues:

If you suspect the shim supply is causing a ghosting or stability issue, see if the problem can be identified by the GRE or FSE stability tests in SPT. Disconnecting the shim supply output cable for before/after testing can be helpful. To isolate to a single output module if one channel has gone bad, you can pull the output module board for that channel pair and still run the shim supply. For instance, you can pull the left hand module which is the Z2/Z3 channels and run the test again to see if the problem goes away. This may indicate an individual module problem or just tell you which channels seem to be involved in the problem. The shim filter has large ferites installed in the equipment room side to filter out noise from the ACGD gradients that were found to cause instabilities in shim output channels. If there are any grounding issues, these problems can still occur. Check all grounding to the pen panel to isolate out this cause. If the shim cables are laying in contact with the gradient cabling, this situation will increase the chances of induced currents on the shim cable that the shim supply will respond to in order to try to maintain a constant current output.

5-5 Troubleshooting a phase oscillation in SPT GRE stability plots:

Illustration 5 is a screen capture of a phase oscillation in GRE stability while running SPT. This stability failure was seen on a TRM system before implementation of the ferrite filters in the shim filter on the Penetration Panel. The cause being coupled energy from the ACGD gradient lines using the shim output cable as a low inductance, low resistance return path which caused the output amplifiers in the shim supply to lose current control.

Addition of the ferites and proper ground wires with each of the ACGD gradient cables eliminated this issue. This may still show up if a system does not have proper or tight ground connections from the ACGD to the pen panel or weak/damaged components in a shim output channel.

The signature is shown in the SPT plot in Illustration 5 as a phase oscillation with no issues seen in the magnitude or echo shift plots. The amplitude and frequency may vary. Disconnecting the shim output cable and rerunning the SPT GRE stability test again is the easiest method to isolate this particular failure mode. If the problem disappears, check system grounding at the cabinets and pen panel.

If the shim cable is surrounded by and up against the gradient cabling, it also may help to separate the shim cable from the gradient cable.

Another test is to pull the modules one at a time and power up the shim supply with just 2 modules plugged in. Run SPT GRE stability with each module removed one at a time to isolate a bad channel. Can pick just the single plane that problem is seen on to reduce time spent running SPT.

The plane that the oscillation primarily shows up on appeared to be dependent primarily on proximity of the shim cable to the gradient cable for that axis.

If these actions do not help, then the shim supply may need to be replaced. If output modules are available as FRU's, then first try the Z2/Z3 output module.

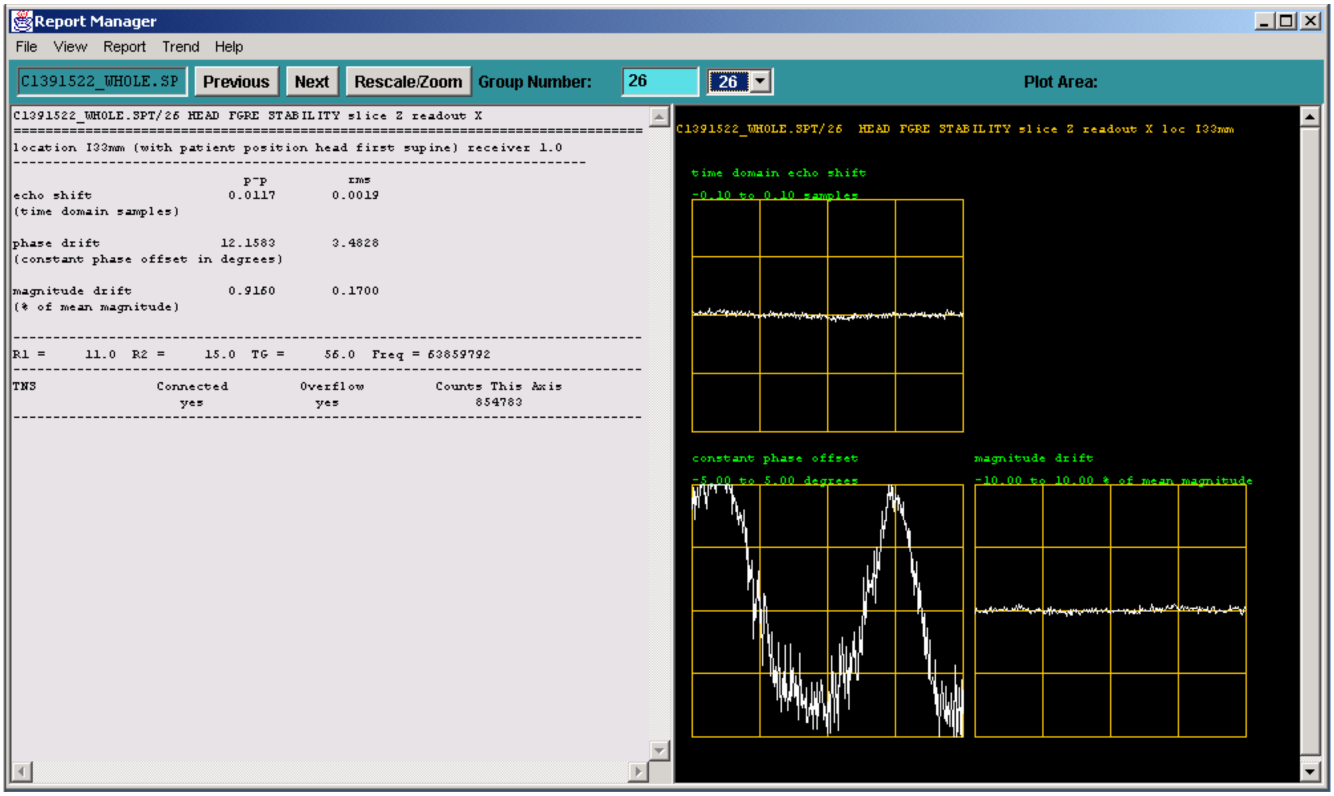


ILLUSTRATION 5
SPT GRE STABILITY

5-6 Shim User Interface does not open

The shim UIF takes awhile to show up after the shim scan completes. This is due to the processing of the data that is necessary in the background. A delay of 5-10 seconds for the 24cm FOV series and a delay of 30-45 seconds for the body 48cm series is typical. During this time there is currently no notification that anything is happening. Do not select scan again until you look into the service log to verify if a problem has occurred. What you will see if the display does not show up is an error stating the "son of recon" failed. Potential reasons:

- **The scan did not complete properly and did not generate a raw data file.** This can be seen by looking in /usr/g/mrraw via "**ls –l P***" and seeing if there is a rawfile Pxxxxx.7 with a time stamp that matches when you last scanned. If not, then rerunning the scan will be necessary.
- **The reference map does not yet exist or is missing for the shim series run.** This can be identifed by looking in **/w/config/ho_shim/reference**. This is where the reference maps are kept. Using "**ls –l**" and looking at the reference.field file link name will tell you what the software is expected to see for the reference map name it is trying to use. Verify that the link is pointing to an existing reference map file name.

- **File corruption.** It may be necessary to delete all P99999.* files since they may have become corrupted. Do this as follows:

  Open a C-shell window and type:
  cd /usr/tmp/ho_autoshim/data <enter>
  rm P99999.* <enter>

Then run the shim scan again and see if the issue is now resolved.

5-7 Problems shimming

One piece of functionality that should not be forgotten to check during any troubleshooting session is the open the “Advanced Screen” and make sure that all the shim coils are enabled. It is possible for someone to disable a coil for some testing/research purpose and leave it disabled. The software does not check and re-enable automatically any of the coils.

Also, as noted in the operator manual, use as large an ROI as realistically possible to give the shim algorithm enough information to differentiate correction coils and apply the best adjustments to the second order shim channels.

6- ADDITIONAL SERVICE INFORMATION:

6-1 3T94 Resisitive Shim tube ZX/ZY harmonic effects swapped

The 3T94 magnet built in England uses the Oxford convention for X and Y which is reversed from the GE convention which has X horizontal. Due to this, if you were to look at the harmonic effects in LVShim while adjusting the High Order Shim Resistive channels ZX and ZY, you would notice that they appear swapped. Since the High Order Shim algorithm uses reference maps to decide what changes to make for in-vivo shimming, it is not affected by the swap in harmonics. This will only be noticed if you look at the harmonics via LVShim or the High Order Shim analysis tool. This swap is also explained in the instructions for the High Order Shim analysis tool which is capable of looking directly at the reference map data.

6-2 Anomolies

If you run a test shim series with the LVShim phantom in the zoom gradient, you may see very dark images in the shim user interface. This will cause the shim calculation to fail since it will only use image data over a certain threshold value as valid data to calculate shim corrections.
This will be true for any image from any shim series that ends up with a very dark image or has severe phase wraps as can be seen by selecting the phase map button in the interface. The severe phase wraps will look just like they would be seen in the LVshim scans, which is black and white stripes adjacent to each other.
The current implementation of the HO Shim process does not know how to handle phase wraps so a bad shim will result from attempting to shim on severely phase wrapped data.

6-3 Errors seen in gesys log

Errors for the resistive shim power supply are currently labeled as “MXA” supply errors. These all refer to the shim power supply installed in the TAC cabinet of a twin system.

6-4 Config changes

The only config file change for the shim power supply option is the “hos_exists” or “high_order_shim_exists” line at the end of the /w/config/prescan_config.cfg file. When set to “1”, the software will expect to see a shim power supply connected to serial expansion port 2 of the host computer. If the shim supply is turned off, you will see failure to communicate to the mxa supply errors in the log when booting the system. Communication to the shim supply can be checked via a C-shell using the “setzero” command. This command sets all 5 shim channels to zero and will display each of the channels as it sets them.

6-5 Reference map naming convention

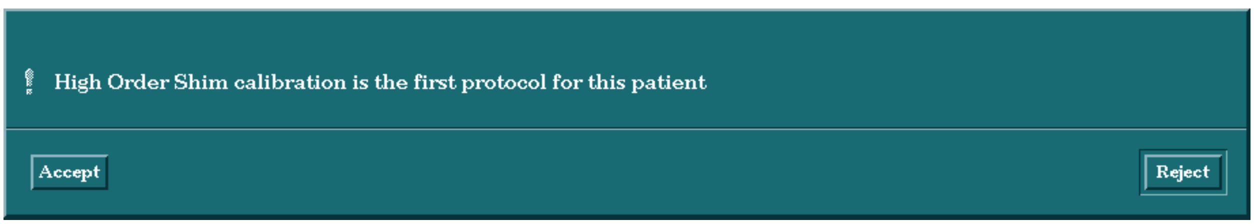
The reference maps are the cal files that are generated by the “makereference” script and are kept in /w/config/ho_shim/reference.
They will all have names such as:
HO_MAP_GRD_W.FOV_24_24_18.RES_64_64_32
The “W” is for Whole GradMode, with other characters possible of “Z” for Zoom and “C” for CRM (3T94 systems).

The FOV is 24cm(X) x 24cm(Y) x 18cm(Z, slice thickness x slices), with a 64x64 pixel resolution in X/Y and 32 slices representing the Z resolution.
You will also see other FOV's depending on system type such as 28cm and 48cm. The larger FOV's such as 48cm will have a larger resolution and twice the slices such as RES_128_128_64.

Each will also have a .params file that is the same name with .params at the end. The params file for each map is a text file that just defines the offset used for each coil when the refmap was generated and a "1" or "0" at the end of each coil line signifying if the coil is currently enabled or disabled by the user via the advanced screen interface option

Notes:

- The shim series will take up an Exam series number but will not place any images in the database. When looking at the browser listing, it will be evident as a missing series. A series may also be missing from an Exam if the series was purposely deleted.
- Use as large an ROI as practical for the anatomy to be shimmed. The larger the area, the better the chance of improving the overall shim. Effectiveness becomes reduced for smaller ROI's due to limited data points for the software to distinguish between shim coils for corrections.
- If you choose to run a shim series as the first series in an exam the following confirmation message will appear:



This message may also appear for the very first shim series after the System software has been started even if the shim series is not the first series. This is a known issue and is currently being investigated.

Don't:

- Don't change any parameter in the shim series other than coil name discussed elsewhere.
- Don't use the GRx box to change slice locations in the shim series.
- Don't use the download button in the Advanced UIF screen more than once per scan. The "Calculate Shim" button can be used as many times as necessary while changing ROI positioning/shape, but only download once.
- Don't Copy/paste any shim series. Always pull in a new copy from the **GE/other** saved protocol HighOrderAutoshim.

6-6 Recommended occasions when HOS calibration should be rerun

- Change in GradMode

- Change in table position
- Change in region to be High Order Shimmed
- When results are not as expected, such as the RMS current value being higher than the last shim iteration during the same shim series.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
A	Nov 29, 2001	S. Merritt P. Kargard	introduction
0	Dec 5, 2001	P. Kargard	Initial version
1	Nov. 13, 2002	S. Merritt	Per SPR 76057